

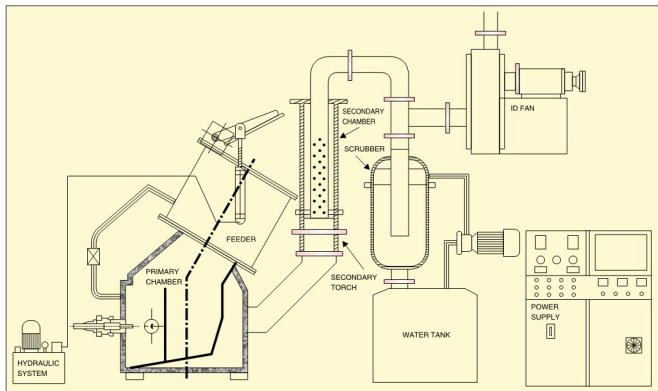


Fig. 10: Schematic of plasma pyrolysis system (courtesy: [www.plasmaindia.com](http://www.plasmaindia.com))

**Startech.** Startech's plasma gasification uses extremely high-energy plasma (at a temperature of 16,649°C, which is three times as hot as the surface of the Sun). The plasma torch inside the containment vessel is directed by an operator to break down whatever material is fed into it. It acts much like contained, continuous lightning, and everything that is fed into the system is broken down into its constituent atoms. Because the plasma torch so thoroughly annihilates anything that is fed into it, it is also an excellent means of dealing with dangerous and hazardous waste. The system is called a closed-loop elemental recycling system.

Startech's plasma converter system is shown in Fig. 9. First, the trash is fed into an auger that shreds it into small pieces. Then the mulch is delivered into the plasma chamber, where the superheated plasma converts it into two byproducts. One is a plasma-converted gas (PCG). Also called syngas, it is composed mostly

of hydrogen and carbon monoxide, and is fed into the adjacent Starcell patented system for conversion into fuel. The other byproduct is molten glass, which can be sold for use in household tiles or road asphalt.

Joseph Longo, CEO of Startech Environmental Corporation, USA, claimed that his plasma converter could turn the most vile and toxic trash into clean energy—and promised to make a relic of the landfill. An article on the same appeared in March 2007 issue of *Popular Science* magazine under the title 'The Prophet of Garbage.' The same can be viewed at '[www.popsci.com/scitech/article/2007-03/prophet-garbage](http://www.popsci.com/scitech/article/2007-03/prophet-garbage)'

Startech's current and future installations include those in Japan and South Carolina, USA. The plasma converter system installed in Japan can process five tonnes per day of hazardous incinerator ash. A joint project with ViTech Enterprises to manufacture and install a 10-tonne-per-day plasma converter facility to destroy out-of-date pharmaceutical products is in progress

in South Carolina, USA.

The company's negotiations with various bodies throughout the world are at various stages of progress.

## Indian scenario

In India, about 30 million tonnes of municipal solid waste and about 4400 million cubic metres of liquid waste are generated annually. The municipal solid waste generation ranges from 0.25 to 0.75 kg/day per capita with an average of 0.45 kg/day per capita. In addition, large quantities of solid and liquid wastes are generated by the industry.

Most of the generated wastes find their way into land and water bodies without proper treatment, causing severe water pollution. These also emit greenhouse gases like methane and carbon dioxide, adding to air pollution.

There exists a potential for generating an estimated 1700 MW of power from the urban and municipal wastes, and about 1000 MW from industrial wastes in the country. The